MODULE 5 MENINGES AND CSF

CNS Meninges and CSF.

THE MENINGES AND CEREBROSPINAL FLUID

Covers the cranial and spinal meninges, the dural reflections, the meningeal spaces, where the spinal cord ends and the lumbar puncture, the ventricles, cerebrospinal fluid and its circulation, and the blood-brain barrier.

BY THE END

1. Name the cranial and spinal meninges and the spaces between them.
2. Identify where the spinal cord ends and the layers a lumbar puncture passes.
3. Describe cerebrospinal fluid, where it is found, and where it is made.
4. Trace the circulation of CSF from production to reabsorption.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Meninges and CSF

drsrennie-stack.github.io/new-build-bio4-solano/meninges-concept-videos.html

An Overview

The brain and spinal cord are soft and irreplaceable, so they are protected four ways: by bone, by the meninges, by a cushioning fluid, and by a barrier that screens the blood.

THE FOUR KINDS OF PROTECTION

Structure	What it is
Protection of the CNS	The brain and spinal cord are shielded by bone, by membranes, by fluid, and by a blood barrier
Bony protection	The cranium encloses the brain; the vertebral column encloses the spinal cord
Meninges	The three connective tissue membranes that wrap the brain and spinal cord
Cerebrospinal fluid	The clear fluid that surrounds, cushions, and floats the central nervous system
Blood-brain barrier	The barrier that controls what passes from the blood into the brain tissue

The Cranial Meninges

Three connective tissue membranes wrap the brain. Their Latin names run from tough mother on the outside to tender mother within. Compare them.

THE THREE LAYERS, OUTERMOST TO INNERMOST

Layer	Position	Description
Dura mater	The outermost layer	A tough membrane with two layers: an outer periosteal layer against the skull and an inner meningeal layer; the two are fused except where they part to form the dural venous sinuses
Arachnoid mater	The middle layer	A delicate, lace-like membrane
Pia mater	The innermost layer	A thin membrane, rich in blood vessels, that clings directly to the surface of the brain

ONE DIFFERENCE FROM THE VERTEBRAL CANAL

Structure	What it is
Cranial epidural space	The periosteal dura is attached directly to the skull, so inside the cranium the epidural space is only a potential space. It opens only if something, such as blood, separates the dura from the skull

The Dural Reflections

In a few places the inner meningeal dura folds inward as a partition that anchors the brain and separates its parts. These folds are the dural reflections, also called the dural extensions.

THE FOLDS OF THE MENINGEAL DURA

Structure	What it is
Dural reflections	Inward folds of the meningeal dura that partition the cranial cavity
Falx cerebri	The fold that dips into the longitudinal fissure, separating the right and left cerebral hemispheres
Falx cerebelli	The fold that separates the right and left cerebellar hemispheres
Tentorium cerebelli	The fold that separates the cerebrum above from the cerebellum below

Where the two dural layers part within these folds, they form the dural venous sinuses. The sinuses collect venous blood from the brain, along with the CSF returned by the arachnoid granulations, and drain it toward the heart. Follow the main drainage route.

1. **Superior sagittal sinus** runs along the top edge of the falx cerebri; receives CSF from the arachnoid granulations
2. **Confluence of sinuses** the meeting point of the major sinuses, at the internal occipital protuberance
3. **Transverse sinuses** the paired sinuses running laterally from the confluence
4. **Sigmoid sinuses** the S-shaped sinuses that curve down toward the jugular foramen
5. **Internal jugular veins** carry the blood out of the skull and back toward the heart

The Spinal Meninges

The same three membranes continue down around the spinal cord, with one important difference: the vertebral canal has a true epidural space, while in the cranium it is only a potential space.

THE MEMBRANES AND ANCHORS AROUND THE CORD

Structure	What it is
Spinal dura mater	A single tough layer around the cord, not fused to bone, unlike the two-layered cranial dura
Spinal arachnoid mater	The middle membrane, continuous with the cranial arachnoid
Spinal pia mater	The innermost membrane, clinging to the surface of the cord
Epidural space	The space between the spinal dura and the vertebral bone, filled with fat and veins; the site of an epidural injection
Denticulate ligaments	Tooth-like extensions of pia mater that anchor the cord to the dura along its length
Filum terminale	A thread of pia mater that extends from the end of the cord and anchors it to the coccyx

The Meningeal Spaces

The gaps around and between the meninges are named spaces. Each one matters in the clinic, whether for an injection, a bleed, or a spinal tap. Compare them.

THE THREE NAMED SPACES

Space	Location	Contents
Epidural space	Between the spinal dura and the vertebral bone	Fat and veins; a real space in the vertebral canal, only a potential space in the cranium
Subdural space	Between the dura mater and the arachnoid mater	A potential space only, with a thin film of fluid
Subarachnoid space	Between the arachnoid mater and the pia mater	Cerebrospinal fluid; this is where CSF is found around the brain and cord

Where the Spinal Cord Ends

The spinal cord is shorter than the vertebral column. It ends high in the lumbar region, and the subarachnoid space continues below it as a pool of CSF.

THE END OF THE CORD AND WHAT LIES BELOW IT

Structure	What it is
Conus medullaris	The tapered end of the spinal cord, at the level of the L1 to L2 intervertebral disc in the adult
Cauda equina	The hair-like bundle of nerve roots that descends past the conus medullaris within the vertebral canal
Filum terminale	The pia mater thread running from the conus medullaris to anchor on the coccyx
Lumbar cistern	The enlarged subarachnoid space below the conus medullaris, filled with CSF and the floating cauda equina

The Lumbar Puncture

A lumbar puncture, or spinal tap, draws CSF from the lumbar cistern. The needle enters in the midline between the L3 to L4 or L4 to L5 spines, safely below the end of the cord. Follow the structures it passes, superficial to deep.

1. **Skin** the epidermis and dermis of the lower back
2. **Subcutaneous fat** the superficial fascia beneath the skin
3. **Supraspinous ligament** the ligament running along the tips of the spinous processes
4. **Interspinous ligament** the ligament between adjacent spinous processes
5. **Ligamentum flavum** the tough elastic ligament joining the laminae of adjacent vertebrae
6. **Epidural space** the fat-filled and vein-filled space inside the vertebral canal
7. **Dura mater** the tough outer meningeal membrane
8. **Arachnoid mater** the delicate middle meningeal membrane
9. **Subarachnoid space** the needle reaches the lumbar cistern, where CSF is withdrawn

The Ventricles of the Brain

The ventricles are connected, fluid-filled chambers deep within the brain, lined by ependymal cells. Cerebrospinal fluid is made in them. Compare them.

THE VENTRICLES AND HOW THEY CONNECT

Ventricle	Location	Connects to
Lateral ventricles	One in each cerebral hemisphere	The third ventricle, through the interventricular foramina of Monro
Third ventricle	In the diencephalon	The fourth ventricle, through the cerebral aqueduct of Sylvius
Fourth ventricle	Between the brainstem and the cerebellum	The central canal and the subarachnoid space, through three apertures
Central canal	Runs the length of the spinal cord	Continuous above with the fourth ventricle

WHAT LINES THEM AND WHAT MAKES THE FLUID

Structure	What it is
Choroid plexus	Networks of capillaries, wrapped in ependymal cells, that produce CSF; found in the lateral, third, and fourth ventricles
Ependymal cells	The glial cells that line the ventricles and central canal and help circulate CSF

Cerebrospinal Fluid

Cerebrospinal fluid is the clear liquid cushion of the central nervous system. It is made, circulated, and reabsorbed continuously, keeping a steady volume.

WHAT CSF IS, WHERE IT IS, AND WHAT IT DOES

Structure	What it is
Cerebrospinal fluid	CSF, the clear, watery fluid that surrounds and fills the central nervous system
Where CSF is found	In the four ventricles, in the central canal of the spinal cord, and in the subarachnoid space around the whole brain and cord, including the lumbar cistern
Site of production	The choroid plexus within the ventricles
Site of reabsorption	The arachnoid granulations, which return CSF to the venous blood of the dural sinuses
Buoyancy	The fluid floats the brain, reducing its effective weight so it does not crush its own base
Protection	The fluid cushions the CNS against jolts and blows
Chemical stability	The fluid carries nutrients, removes wastes, and keeps a stable environment around the neurons

The Circulation of CSF

Cerebrospinal fluid follows a one-way loop, produced in the ventricles and reabsorbed into the blood. The ventricles below hold choroid plexus, where ependymal cells produce CSF. Follow it in order.

1. **Lateral ventricles** hold choroid plexus, where ependymal cells produce CSF; one ventricle sits in each cerebral hemisphere
2. **Interventricular foramina of Monro** CSF passes through these paired openings into the third ventricle
3. **Third ventricle** in the diencephalon; also holds choroid plexus, adding more CSF
4. **Cerebral aqueduct of Sylvius** CSF flows down this channel through the midbrain
5. **Fourth ventricle** between the pons and the cerebellum; also holds choroid plexus
6. **Apertures of the fourth ventricle** CSF exits through the two lateral apertures of Luschka and the median aperture of Magendie

7. **Subarachnoid space** CSF bathes the whole surface of the brain and spinal cord; some passes into the central canal
8. **Arachnoid granulations** CSF is reabsorbed here into the venous blood of the superior sagittal sinus, completing the loop

KEY POINT

CSF is made in the ventricles but reabsorbed outside them, at the arachnoid granulations in the superior sagittal sinus. Anything that blocks the narrow points on that route, the foramina, the aqueduct, or the apertures, backs the fluid up behind it.

The Blood-Brain Barrier

The blood-brain barrier is the structural seal that keeps brain tissue separate from the changing chemistry of the blood. It is built from three components.

THE BARRIER AND ITS THREE COMPONENTS

Structure	What it is
Blood-brain barrier	The BBB, a selective barrier between the blood and the brain tissue
Capillary endothelial cells	The cells lining brain capillaries, joined by tight junctions and lacking the pores of ordinary capillaries
Basement membrane	The supporting layer that wraps the capillary endothelium
Astrocyte end-feet	The foot-like projections of astrocytes that envelop the capillary and help control the barrier

Disorders of the Meninges and CSF

Compare the common disorders by the structure each one affects.

THE DISORDERS AND THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Meningitis	The meninges	Inflammation of the meninges, usually from infection; diagnosed by testing CSF drawn at a lumbar puncture
Hydrocephalus	The ventricles	A buildup of CSF that enlarges the ventricles and raises intracranial pressure
Epidural hematoma	The epidural space	Bleeding between the dura mater and the skull
Subdural hematoma	The subdural space	Bleeding between the dura mater and the arachnoid mater
Subarachnoid hemorrhage	The subarachnoid space	Bleeding into the CSF-filled space around the brain

THE CORD ENDS, BUT THE FLUID DOES NOT

The spinal cord stops at the **conus medullaris** near the L1 to L2 disc, yet the subarachnoid space runs lower still, as the **lumbar cistern**. That mismatch is the whole basis of the lumbar puncture: a needle placed between the L3 to L4 or L4 to L5 spines reaches a pool of CSF with only the floating **cauda equina** there to drift aside.

Knowing the layers in order is what makes the tap safe. The needle crosses skin, fat, three ligaments, the **epidural space**, and then the dura and arachnoid before CSF appears. A clinician feels the firm **ligamentum flavum** give way as a landmark.

The named spaces also tell a bleed apart. An **epidural hematoma** sits outside the dura, a **subdural hematoma** just beneath it, and a **subarachnoid hemorrhage** mixes into the CSF itself. Each location is found, and treated, differently.